COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	08 - 28 - 26	**DATE SUBMITTED:**	09 - 04 - 26
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HOA NO. _6__ Searching Algorithm

1. PROCEDURE OUTPUT

6.1. Data Generated and Observations

Screenshot

```
1   #include <iostream>
2   #include <cstdlib> //for generating random int
3   #include <time.h> //will be used for our seeding function
4   #include "searchingAlgo.h"
5   const int max_size = 50;
6   int main(){
7       int dataset[max_size];
8       srand(time(0));
9       for(int i = 0; i <max_size; i++){
10          dataset[i] = rand();
11          | std::cout<<dataset[i]<< " ";
12      }
```

Observations

Before testing the search functions, initializing a random array of 50 numbers is created as test data, this is generated and stored using the rand() function. This is for the linear and binary search algorithms have a initialization of dataset.

6.2a. Linear Search for Arrays

Code

```
template<typename T>
void linearSearchAR(int size, T dataFind, T array[]){
    //while loop or for loop

    for(int j = 0; j < size; j++){
        if(dataFind == array[j]){
            std::cout<<"Searching is successful\n";
            return;
        }
        j++;
    }
    std::cout<<"Searching is unsuccessful\n";

}
```

Output

```
C:\Users\Paramount Laptop 5
23299 1855 19907 3453 30123 25468 4522 8243 27024 11773 1
6599 25299 15474 17232 15416 3821 27827 26398 15447 12330
 30568 505 10266 28720 1200 27732 17324 22982 11845 27347
 19387 22329 5200 16326 20708 28670 13442 30192 26475 245
6 14176 18572 32560 12221 10206 19602 30686 10164 18736 1
6182
What number to find: 23299
Searching is successful
Data D was found successfully
The data was not found
Searching is unsuccessful

--------------------------------
Process exited after 9.675 seconds with return value 0
Press any key to continue . . .
```

Observations

I have observed in the Linear Search for Arrays is where when the number input by the user is on the dataset given, it will print Searching is successful, it checks the whole list and prints after, so when the user input is a random number not to be seen or it is not generated in the random array, it will print " Searching is unsuccessful".

6.2b. Binary Search for Arrays

Code

```
template <typename T>
void linearSearchLL(Node<T> *nodes, T dataFind){
    //repeat the loop until the current node is not NULL
    while(nodes != nullptr){
        //check if the item = current node data
        if(nodes->data == dataFind){
            std::cout<< "Data "<< nodes->data <<" was found successfully\n";
            return;
        }
        nodes = nodes->next;
    }
    std::cout<<"Searching is unsuccessful\n";
}
```

Output

```
C:\Users\Paramount Laptop 5
23299 1855 19907 3453 30123 25468 4522 8243 27024 11773 1
6599 25299 15474 17232 15416 3821 27827 26398 15447 12330
 30568 505 10266 28720 1200 27732 17324 22982 11845 27347
 19387 22329 5200 16326 20708 28670 13442 30192 26475 245
6 14176 18572 32560 12221 10206 19602 30686 10164 18736 1
6182
What number to find: 23299
Searching is successful
Data D was found successfully
The data was not found
Searching is unsuccessful

--------------------------------
Process exited after 9.675 seconds with return value 0
Press any key to continue . . .
```

Observations

Data D was found successful because in the main.cpp file, there is a given name (JIMLORD) where it checks the given, so if the node is found like D, it will print successful, but if, for example, I tried to finding X, it will print Searching is unsuccessful because it check per node until it reaches the null, then prints.

6.3a. Binary Search for Arrays

Code

```
template <typename T>
void binarySearchArr(T data[], int arrSize, T dataFind){
    int low = 0, up = arrSize - 1 ; int middle;
    while(low <= up){
        middle = (int)((low+up/2));
        if(dataFind == data[middle]){
            std::cout<<"The data was found\n";
            return;
        }
        else if(dataFind < data[middle]){
            up = middle - 1;
        }
        else{
            low = middle + 1;
        }
    }
    std::cout<<"The data was not found\n";
}
```

Output

```
C:\Users\Paramount Laptop 5
23299 1855 19907 3453 30123 25468 4522 8243 27024 11773 1
6599 25299 15474 17232 15416 3821 27827 26398 15447 12330
 30568 505 10266 28720 1200 27732 17324 22982 11845 27347
 19387 22329 5200 16326 20708 28670 13442 30192 26475 245
6 14176 18572 32560 12221 10206 19602 30686 10164 18736 1
6182
What number to find: 23299
Searching is successful
Data D was found successfully
The data was not found
Searching is unsuccessful

--------------------------------
Process exited after 9.675 seconds with return value 0
Press any key to continue . . .
```

Observations

The data was not found because in the main.cpp file, the code is searching for the value of 11 inside the newark, but the initialized array is only at 1-10, so when the loop searched the whole array and exited with not finding the 11 because its does not exit, therefore, printing “The data was not found”.

6.3b. Binary Search for Linked List

Code

```
77  template <typename T>
78  void binarySearchLL(Node<T>* start, T dataFind){
79      Node<T>* last = nullptr;
80      while(start != last){
81          Node<T>* middle = getMiddle(start, last);
82          if(middle->data == dataFind){
83              std::cout<<"Searching is successful\n";
84              return;
85          }
86          else if(middle->data > dataFind){
87              last = middle;
88          }
89          else{
90              start = middle->next;
91          }
92      }
93      std::cout<<"Searching is unsuccessful\n";
94  }
95  #endif
```

Output

```
C:\Users\Paramount Laptop 5
23299 1855 19907 3453 30123 25468 4522 8243 27024 11773 1
6599 25299 15474 17232 15416 3821 27827 26398 15447 12330
 30568 505 10266 28720 1200 27732 17324 22982 11845 27347
 19387 22329 5200 16326 20708 28670 13442 30192 26475 245
6 14176 18572 32560 12221 10206 19602 30686 10164 18736 1
6182
What number to find: 23299
Searching is successful
Data D was found successfully
The data was not found
Searching is unsuccessful

--------------------------------
Process exited after 9.675 seconds with return value 0
Press any key to continue . . .
```

Observations

Searching is unsuccessful because in the main.cpp file, the binarySearchLL is finding for 18, since the linked list is only at 10-16 only, 18 doesn't exist, so after checking the list, it prints "Searching is unsuccessful" but if the input in the source code in main.cpp is change to the range of 10-16 it will print successful.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Problem 1:	
Code	main.cpp 2searchingAlgo.h node.h

```cpp
#include <iostream>
#include <cstdlib>
#include <time.h>
#include "2searchingAlgo.h"

const int max_size = 50;

int main(){
    int dataset[max_size];
    srand(time(0));
    for(int i = 0; i < max_size; i++){
        dataset[i] = rand();
        std::cout << dataset[i] << " ";
    }
    std::cout << std::endl;

    std::cout << "What number to find: ";
    int choice;
    std::cin >> choice;
    linearSearchAR(max_size, choice, dataset);

    Node<int> *name1 = new_node(15);
    Node<int> *name2 = new_node(18);
    Node<int> *name3 = new_node(2);
    Node<int> *name4 = new_node(19);
    Node<int> *name5 = new_node(18);
    Node<int> *name6 = new_node(0);
    Node<int> *name7 = new_node(8);
    Node<int> *name8 = new_node(14);
    Node<int> *name9 = new_node(19);
    Node<int> *name10 = new_node(14);

```

```
33      name1->next = name2;
34      name2->next = name3;
35      name3->next = name4;
36      name4->next = name5;
37      name5->next = name6;
38      name6->next = name7;
39      name7->next = name8;
40      name8->next = name9;
41      name9->next = name10;
42
43
44      linearSearchLL(name1, 18);
45
46
47      int newArr[10] = {15, 18, 2, 19, 18, 0, 8, 14, 19, 14};
48      linearSearchAR(10, 18, newArr);
49
50
51      Node<int> *Name1 = new_node(10);
52      Node<int> *Name2 = new_node(11);
53      Node<int> *Name3 = new_node(12);
54      Node<int> *Name4 = new_node(13);
55      Node<int> *Name5 = new_node(14);
56      Node<int> *Name6 = new_node(15);
57      Node<int> *Name7 = new_node(16);
58
59
60      Name1->next = Name2;
61      Name2->next = Name3;
62      Name3->next = Name4;
63      Name4->next = Name5;
64      Name5->next = Name6;
65      Name6->next = Name7;
66
67      binarySearchLL(Name1, 18);
68
69      return 0;
70  }
```

main.cpp node.h 2searchingAlgo.h

```
13  void linearSearchAR(int size, T dataFind, T array[]){
14      int comparisons = 0;
15      int foundCount = 0;
16      //while loop or for loop
17
18      for(int j = 0; j < size; j++){
19          comparisons++; //to increment in every check
20          if(dataFind == array[j]){
21              foundCount++;
22          }
23      }
24      if(foundCount > 0 ){
25          std::cout<<"Searching is successful! The Total of Comparison: " << comparisons << "\n";
26      }else{
27          std::cout<<"Searching is unsuccessful\n";
28      }
29  }
```

main.cpp | 2searchingAlgo.h | node.h

```
31  template <typename T>
32  void linearSearchLL(Node<T> *nodes, T dataFind){
33      int comparisons = 0;
34      int foundCount = 0;
35
36      while(nodes != nullptr){
37          comparisons++;
38          if(nodes->data == dataFind){
39              foundCount++;
40          }
41          nodes = nodes->next;
42      }
43      if(foundCount > 0){
44          std::cout << "Linked List Search Successful! Total Comparisons: " << comparisons << "\n";
45      }else{
46          std::cout<<"Searching is unsuccessful\n";
47      }
48  }
49
```

Output

```
C:\Users\Paramount Laptop 5
22220 19938 17350 26133 29413 12849 16451 31384 1197 18214 31167 24408 13
329 27719 2995 21432 19838 23712 9226 28871 23888 11368 6708 13321 8078 1
8913 31862 10107 6100 1408 1924 19153 26000 25934 15441 8706 10353 18318
2086 3334 2333 6304 13042 19678 14883 8229 2017 22998 1328 10596
What number to find: 90
Searching is unsuccessful
Linked List Search Successful! Total Comparisons: 10
Array Linear Searching is successful! The Total of Comparison: 10
Searching is unsuccessful

--------------------------------
Process exited after 3.037 seconds with return value 0
Press any key to continue . . .
```

Question to Answer:

In this problem one, I retained the original tasks before the supplementary. So the ones I needed to do is modify the linked list and array of linear search, where I changes the list first then I put two integers where it will be used if the nodes are not equal to null, that means its not in the tail. The code will check the list one by one, to know how many comparisons it will print, since the list consists of 10 elements, the comparisons will be compared ten times, that is why 10 is printed in total comparisons.

Problem 2:

Code

searchingalgo.h main.cpp node.h

```
20
21      std::cout << "Occurrences in dataset: " << linearSearchAR(max_size, choice, dataset) << "\n\n";
```

searchingalgo.h main.cpp node.h

```
44
45      linearSearchLL(name1, 18);
46
47
48      int newArr[10] = {15, 18, 2, 19, 18, 0, 8, 14, 19, 14};
49      linearSearchAR(10, 18, newArr);
50
```

searchingalgo.h main.cpp node.h

```
12  template<typename T>
13  int linearSearchAR(int size, T dataFind, T array[]){
14      int matchCount = 0;
15      //while loop or for loop
16
17      for(int j = 0; j < size; j++){
18          if(array[j] == dataFind){
19              matchCount++;
20          }
21      }
22      std::cout << "Element " << dataFind << " appears " << matchCount << " times in the array.\n";
23
24  }
25
26  template <typename T>
27  int linearSearchLL(Node<T> *nodes, T dataFind){
28      int matchCount   = 0;
29
30      while(nodes != nullptr){
31          if(nodes->data == dataFind){
32              matchCount++;
33          }
34          nodes = nodes->next;
35      }
36      std::cout << "Element " << dataFind << " appears " << matchCount << " times in the array.\n";
37
38  }
```

Output

C:\Users\Paramount Laptop 5

```
2602 25288 7346 9878 6446 11020 10548 356 30477 30934 29401 20392 15844
30115 21947 13160 9399 25911 3101 11560 23557 24546 4001 18515 6303 513
6843 26333 16570 13511 24165 28711 28482 29758 16404 26403 16753 13960 1
3291 11192 8865 22635 1839 27496 2654 13069 12028 1328 24768 16123
What number to find: 2
Element 2 appears 0 times in the array.
Occurrences in dataset: 4745728

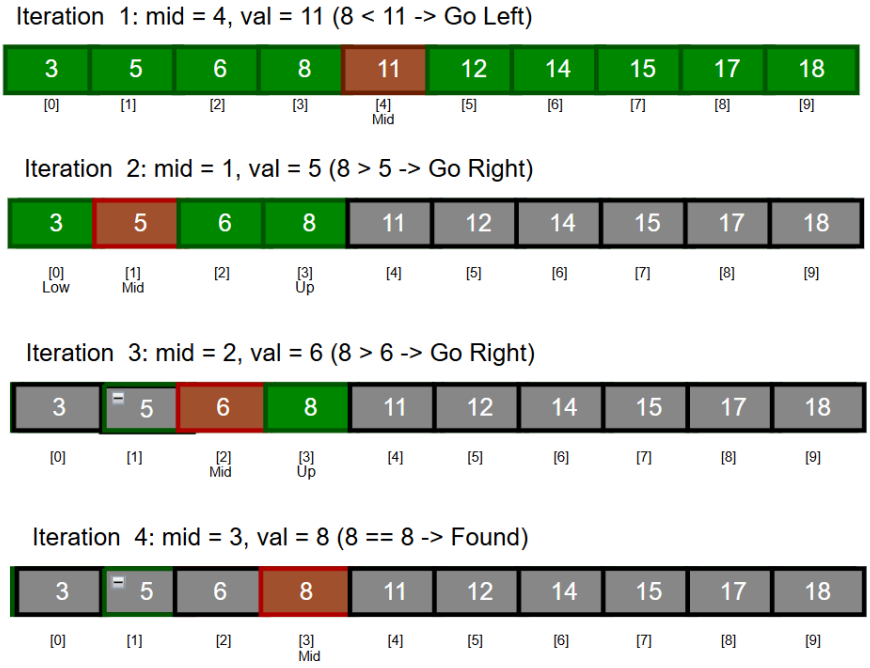
Element 18 appears 2 times in the array.
Element 18 appears 2 times in the array.
Searching is unsuccessful

--------------------------------
Process exited after 2.21 seconds with return value 0
Press any key to continue . . .
```

Observations	In this problem two, I retained the original tasks before the supplementary like in problem on. I focused on the number of 18, I could've used user input where it will be finding the number that will be input to, but I stayed on the easy task, because I was curious how it will end up. The list retained as is because it will be basing on that. The for loop says that if it is less than the size of the number given it will search how many times it appeared in the list. In this case element 18 appeared two times in the list.
Problem 3:	

Code	searchingalgo.h main.cpp node.h
	`int newArr[10] = {3, 5, 6, 8, 11, 12, 14, 15, 17, 18};`
	`binarySearchArr(newArr, 10, 8 );`
	
	searchingalgo.h main.cpp node.h
	`void binarySearchArr(T data[], int arrSize, T dataFind){`
	`int low = 0, up = arrSize - 1 ; int middle; int iteration = 1;`
	`while(low <= up){`
	`middle = (int)((low+up/2));`
	`std::cout << "Iteration " << iteration++ << ": Low=" << low << ", Up=" << up << ", Mid=" << middle << ", Value=" << data[middle] << "\n";`
	`if(dataFind == data[middle]){`
	`std::cout<<"The data " << dataFind << " was found at index " << middle << "\n";`
	`return;`
	`}`
	`else if(dataFind < data[middle]){`
	`up = middle - 1;`
	`}`
	`else{`
	`low = middle + 1;`
	`}`
	`}`
	`std::cout<<"The data was not found\n";`
	`}`

Output	C:\Users\Paramount Laptop 5 9198 11884 15405 22828 1438 13158 6243 18268 26066 30909 25543 11062 8053 3 0334 23491 7111 16241 20415 31138 17794 14205 2600 23551 32226 13110 24243 25725 14928 17266 1967 23993 28230 21457 12299 15741 5301 7829 26577 22327 26788 4787 16881 4777 4378 24617 32576 31457 17471 7827 27328 What number to find: 2 Element 2 appears 0 times in the array. Occurrences in dataset: 4749824 Element 18 appears 2 times in the array. Iteration 1: Low=0, Up=9, Mid=4, Value=11 Iteration 2: Low=0, Up=3, Mid=1, Value=5 Iteration 3: Low=2, Up=3, Mid=3, Value=8 The data 8 was found at index 3 Searching is unsuccessful -------------------------------- Process exited after 1.875 seconds with return value 0 Press any key to continue . . .



Observations

In this part I had to use gemini to understand what to do, so first I changed the list, then to see what's happening, I tried to do the diagram, so by the time I got it I tried to code and modify what I need to do. Using what gemini gave me as a reference, I did a step by step search then I printed what was at low, up, and mid point, so I given picked 8 to be looked upon in the array, where it was found in index 3.

Problem 4:

Code

searchingalgo.h | main.cpp | node.h

```
48      int newArr[10] = {3, 5, 6, 8, 11, 12, 14, 15, 17, 18};
49      binarySearchRC(newArr, 0, 9, 8 );
50
```

searchingalgo.h | main.cpp | node.h

```
39
40  template <typename T>
41  void binarySearchRC(T data[], int lowRC, int upRC, T dataFind){
42
43      while(lowRC > upRC){
44          std::cout<<"The data was not found\n";
45          return;
46      }
47      int middle = (int)((lowRC+upRC/2));
48
49      if(dataFind == data[middle]){
50          std::cout<<"The data " << dataFind << " was found at index " << middle << "\n" ;
51          return;
52      }
53      else if(dataFind < data[middle]){
54          binarySearchRC(data, lowRC, middle - 1, dataFind);
55      }
56      else{
57          binarySearchRC(data, middle + 1, upRC, dataFind);
58      }
59
60
61  }
```

Output

C:\Users\Paramount Laptop 5

```
4109 20337 11194 2675 2747 3833 9138 12062 26177 210
07 17074 23954 27596 29954 3172 289 13935 20324 1628
3 19618 8122 15958 18313 14031 23183 9222 19762 5612
 12337 117 32032 7850 5022 1918 21848 19018 32543 22
548 3982 26629 20509 14539 31637 23980 5430 32153 24
089 23436 656 31454
What number to find: 98
Element 98 appears 0 times in the array.
Occurrences in dataset: 4749824

Element 18 appears 2 times in the array.
The data 8 was found at index 3
Searching is unsuccessful

--------------------------------
Process exited after 2.235 seconds with return value
 0
Press any key to continue . . .
```

Observations	In problem four, i replaced while loop with recursive method, this method is a sequence like it starts with 3 and adds 2 to the previous number, so using low and up as directly function arguments, I firstly put the condition where if low is bigger than up it the code will exit immediately. So if the the data is equal to the middle, it will be defined so first low will be added to up and the answer will be divided to two, at this way its much easier to find where the element is. In this activity I wanted to know where 8 is, and it printed that 8 was at index 3.

Conclusion:

In my conclusion in this activity are the ones that I have learned and the things I don't understand are firstly, initialize the things that are needed to work, to know those things, it can be founded at the errors or the the ones we have learned. Secondly , know the difference between array and linked list, what can be used in the the most situation, array are in fixed size, linked list are dynamic in size means they can grow or shrink. Thirdly, I need to explore more and have time to code, because there's a lot of ways to find a number, which in this activity there are nine methods and I get how but I have get to understand the code word for word to individually get the point. Lastly, this way of understanding the code where it stays on the same topic and uses the code and modifying it works far more better for me, because I get to understand why and because I have to change it for it to work, the error analysis is very useful that the code itself.

3vb . ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Explanation of my idea of the code**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **The idea is original but it need to be double check for correct idea and thinking.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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